



Generating impact in research

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Mario is a coach, facilitator and learning & development consultant. Before becoming a coach, he worked for the Royal Society of Chemistry.

Mario has extensive experience of leading teams in higher education policy, regulation and industry policy. He ran programmes in technology policy, in supporting small companies and at the academic – business interface. Mario has a PhD in chemistry and is a Chartered Scientist.

He trained as a coach and is a certified member of the International Coach Federation. He has worked with researchers across many sectors throughout his career.

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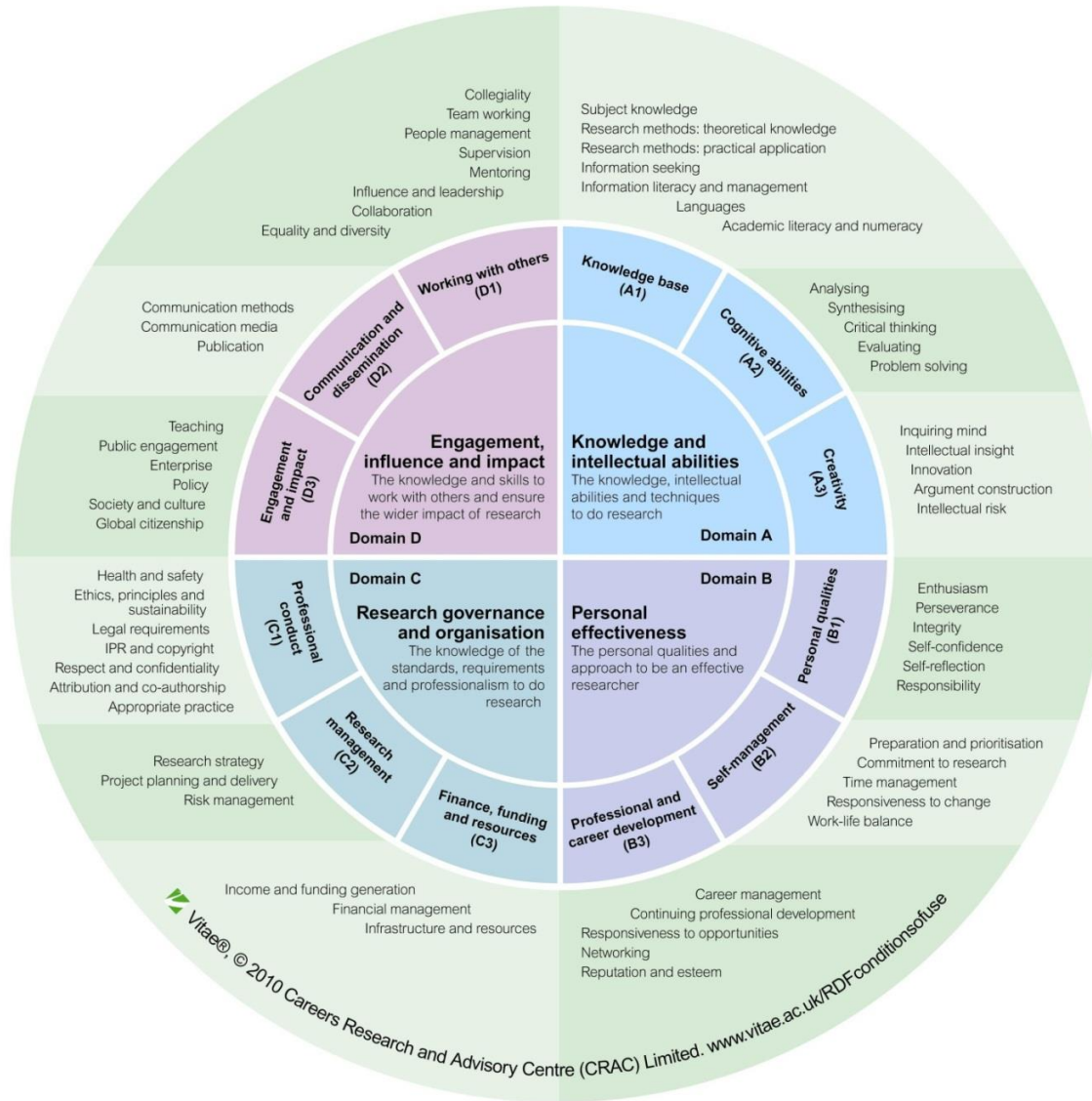
Aim of the webinar



The aim of the webinar is to develop your knowledge and understanding of:

- What is research impact;
- Examples of different types of research impact;
- How to create and evaluate research impact.

Vitae Researcher Development Framework



- Framework of the knowledge, behaviour and attributes of successful researchers
- Enables self-assessment of strengths and areas for further development
- Common language for researchers capabilities

Research impact

What is research impact?

A benefit or positive change that takes place outside academia because of your research. The demonstrable contribution that excellent social and economic research makes to society and the economy.

- Research Councils UK

The impact of innovation can be societal, research, environmental, technical, commercial, educational, or anything that delivers a benefit to someone or addresses a need.

- Horizon 2020

Impact can be defined as demonstrable benefits to individuals, organisations and society that could not have been possible without new knowledge arising from your research

- Reed, 2016

An effect on, change or benefit to the economy, society, culture, public policy or services, health, the environment or quality of life, beyond academia.

- REF

What is research impact?

‘Impact is the demonstrable contribution that excellent research makes to society and the economy.

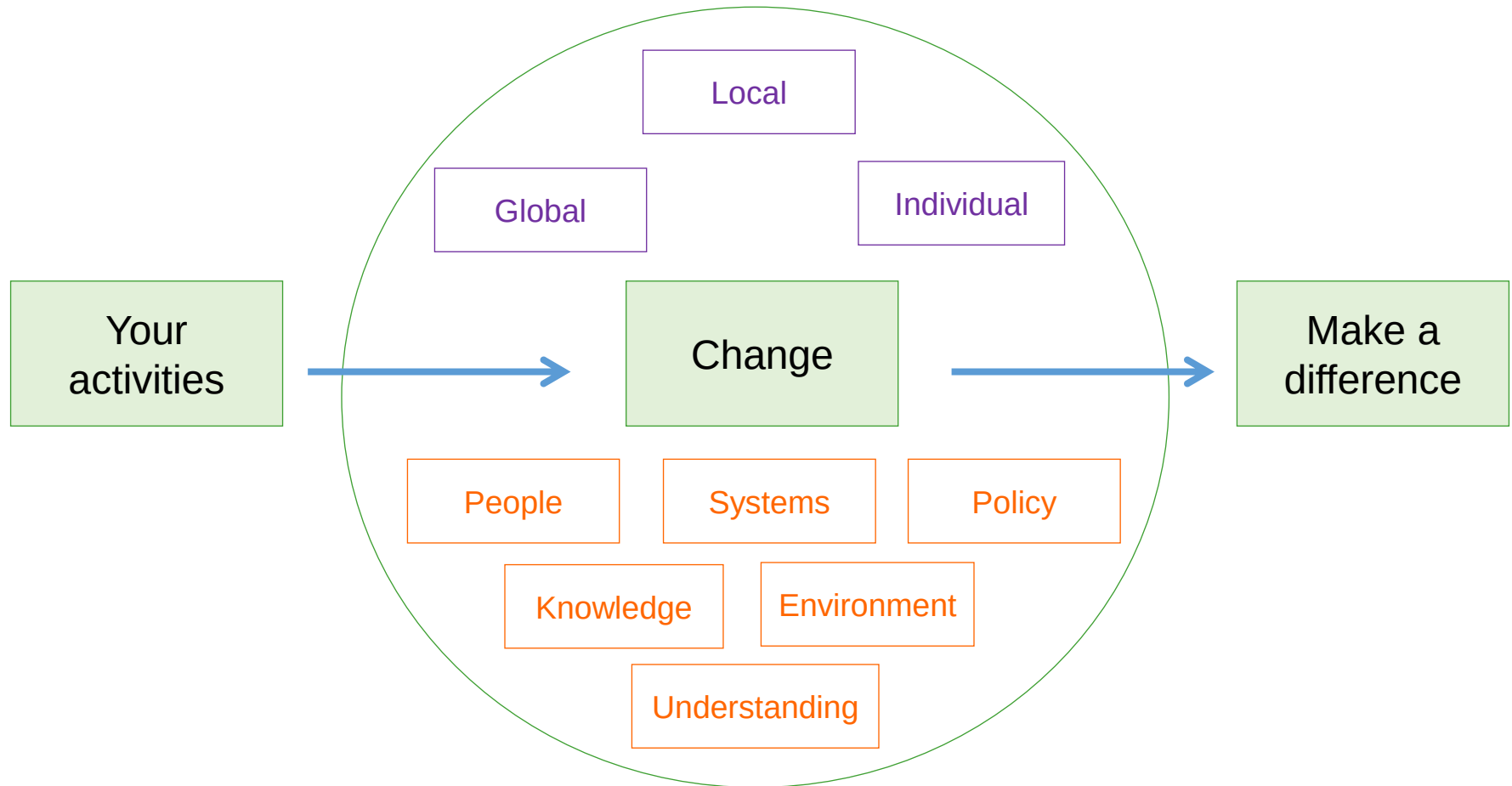
This occurs in many ways including:

- creating and sharing new knowledge and innovation;
- inventing ground-breaking new products, companies and jobs;
- developing new and improving existing public services and policy; and enhancing quality of life and health.’

<https://www.ukri.org/innovation/excellence-with-impact/>

ESRC Research Impact Toolkit: <https://esrc.ukri.org/research/impact-toolkit/>

What is research impact?



Adapted Source: Ref. Colin Chandler, Achieving Impact in Research, SAGE, 2014

Types of impact

Academic impact

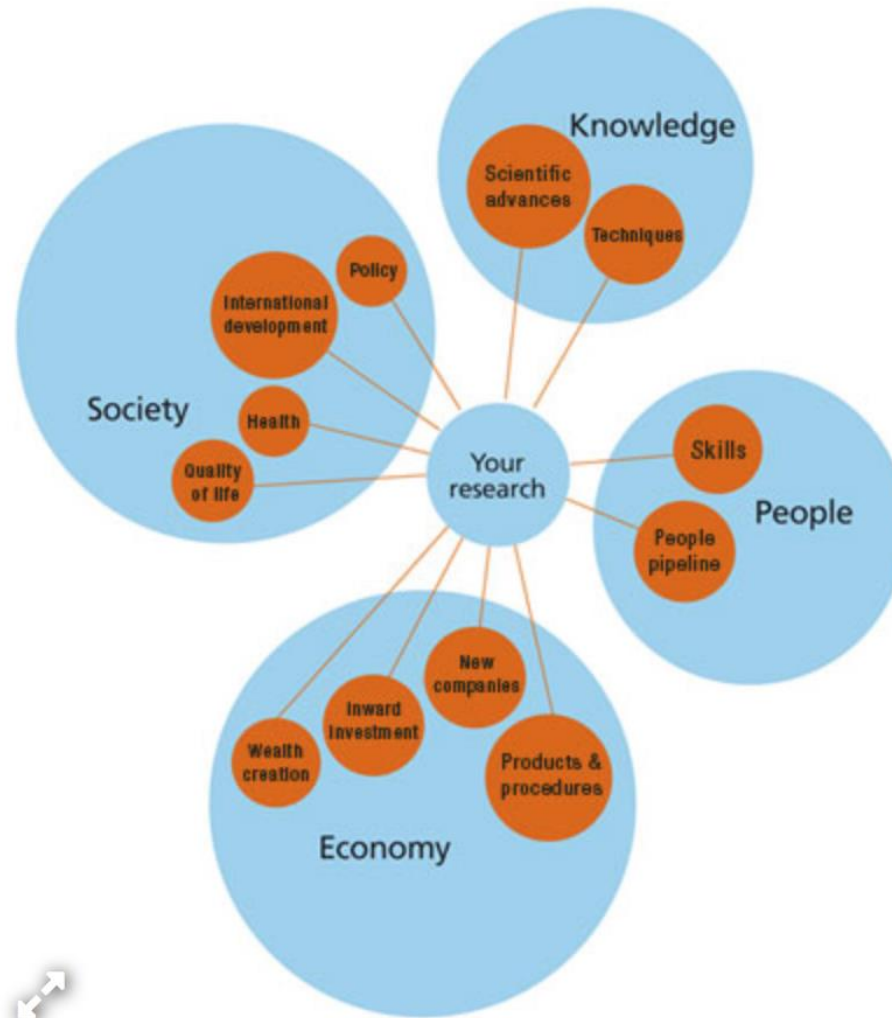
Is the demonstrable contribution that excellent research makes in shifting understanding and advancing method, theory and application across and within disciplines.

Economic and societal impact

is the demonstrable contribution that excellent research makes to society and the economy, of benefit to individuals, organisations and nations.

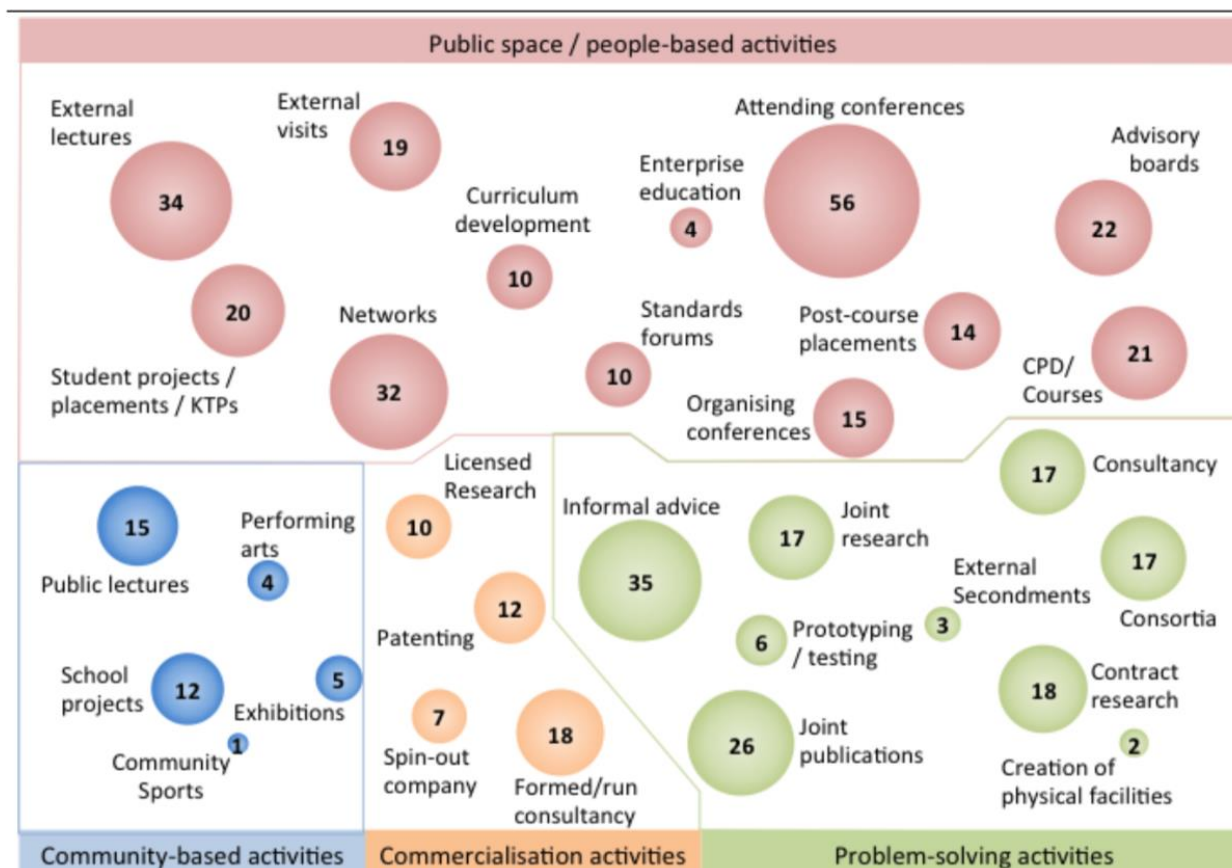
<https://esrc.ukri.org/research/impact-toolkit/what-is-impact/>

Variety of impacts that could be achieved through research



EPSRC Pathways to Impact (2019)

Academic engagement in different types of knowledge exchange mechanisms



Number in bubbles is the % of academics engaging at least three times over the past three years in that mechanism, or at least once in the past three years for commercialisation activities.

Source: PACEC/CBR (2009b) *The Evolution of the Infrastructure of the Knowledge Exchange System*, a report to HEFCE

The African Academy of Sciences – impact across Africa

Identify and support **rising research leaders** to stay
and build their careers in Africa

Setting the research agenda for Africa's most pressing concerns



Postdoctoral fellowship programmes supporting >250 early career researchers

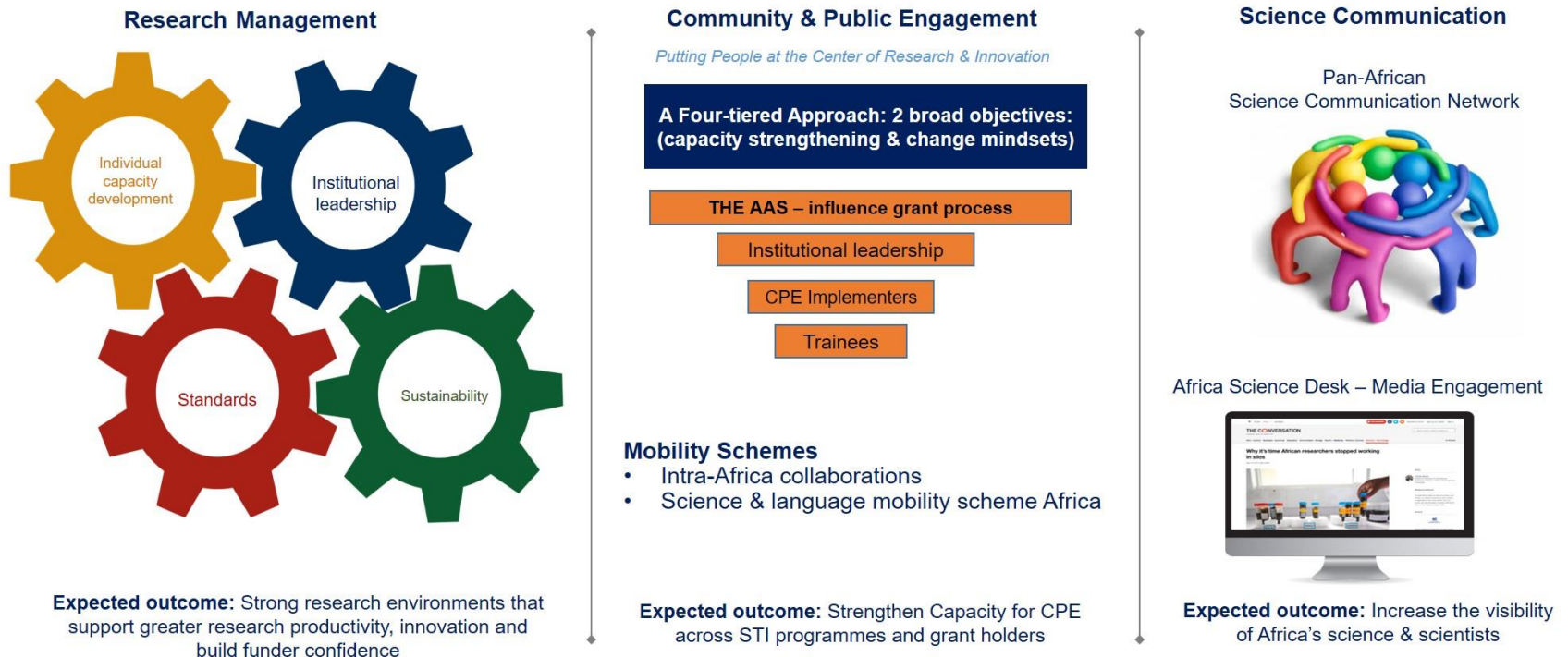
Postdoc Programmes	Grantees	Science Areas	African Region	Partners
Affiliates	87	Natural sciences	C,E,W,N,S	The AAS Endowment Fund & Kevin Marsh thru Al-Sumait 2015 Health Prize
APTI	10	Biomedical sciences	E,W,N	NIH National Institutes of Health BILL & MELINDA GATES foundation
CIRCLE	96	Climate sciences	E,W,S	UKaid
CR4D	21	Climate sciences	C,E,W,S	United Nations Economic Commission for Africa WISER UKaid CR4D
FLAIR	30	Natural sciences	C,E,W,S	THE ROYAL SOCIETY
RISE	7	Natural sciences	E,W,S	<i>Lamarc</i>

African Postdoctoral Training Initiative (APTI), Climate Impacts Research Capacity & Leadership Enhancement (CIRCLE), Climate Research for Development (CR4D), Future Leaders – African Independent Research Fellowship (FLAIR) & Regional Initiative in Science & Education (RISE)

<https://www.aasciences.africa/aesa/activities>

The African Academy of Sciences – impact across Africa

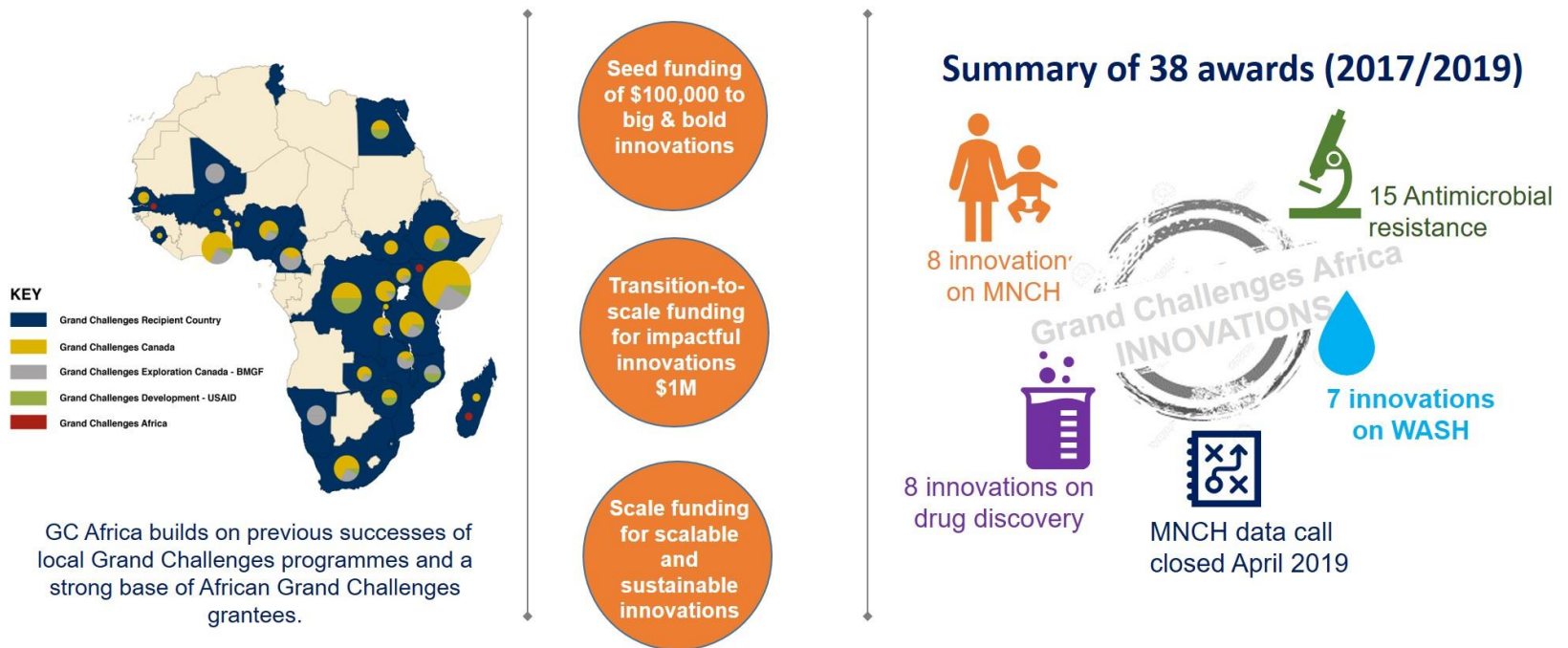
Targeting **critical gaps** in the research landscape



<https://www.aasciences.africa/aesa/activities>

The African Academy of Sciences – impact across Africa

Support the development of an **innovation** and **science-driven**
entrepreneurial culture



<https://www.aasciences.africa/aesa/activities>

Research Councils UK Expectations for Societal and Economic Impact (1)



Those who receive funding should:

- **‘Engage actively with the public** at both the local and national levels about their research and its broader implications.
- **Identify potential benefits and beneficiaries from the outset**, and through the full life cycle of the project(s).
- **Maintain professional networks** that extend beyond their own discipline and research community.
- **Publish results widely** – considering the academics, user and public audiences for research outcomes.’

<https://epsrc.ukri.org/funding/managing/publication/>

Research Councils UK Expectations for Societal and Economic Impact (2)



Those who receive funding should:

- **'Exploit results** where appropriate, in order to secure social and economic return to the UK.
- **Manage collaborations professionally**, in order to secure maximum impact without restricting the future progression of research.
- Ensure that research staff and students **develop research, vocational and entrepreneurial skills** that are matched to the demands of their future career paths.
- Take responsibility for the duration, **management and exploitation of data** for future use.'

<https://epsrc.ukri.org/funding/managing/publication/>

Types of research impact

Typical outcomes for impact

- Publications
- Collaborations
- Further funding
- **Engagement activities**
- **Influence on policy**
- Research tools & methods
- Research databases & models
- IP and licensing
- Medical products & clinical trials
- Artistic & creative products
- Software & technical products
- **Creating an enterprise**
- Awards and recognition

Creating an enterprise

Version:



Business Model Canvas – nine elements



- | | |
|---------------------------|-------------------|
| 1. Customer segments | 6. Key resources |
| 2. Value proposition | 7. Key activities |
| 3. Channels | 8. Key partners |
| 4. Customer relationships | 9. Cost structure |
| 5. Revenue streams | |

‘Business Model Generation’, A. Osterwalder & Y. Pigneur, John Wiley and Sons (2010)

Stratgyzer YouTube Channel

<https://www.youtube.com/channel/UCPVreN9tVxFY2RgWeENShpg>

Value Proposition / USP

“The promise of value to be delivered and the belief from the customer that value will be experienced.”

“Strategy is based on a differentiated customer value proposition. Satisfying customers is the source of sustainable value creation.” *Kaplan*

Value Proposition:

What is your value proposition?

Uniqueness:

What's unique about it?

- Relevancy to customers - **what problem does it solve?**
- Quantify the value - faster, bigger, smaller, easier.
- Place priority on your point of difference – why is your solution better than alternatives?

Customers

A customer (client, buyer, or purchaser) is the recipient of a service or product obtained from a seller, vendor, or supplier for a monetary or other valuable consideration.

Three types:

- An Intermediate Customer or trade customer: Dealer that purchases goods for re-sale.
- An Ultimate Customer: Who does not in turn re-sell the things bought, but either passes them to the consumer or actually is the consumer.
- Internal Customers: Anyone who works with/for you to develop the business.

Customers:

Who are your Customers?

Customer Segmentation: age, gender, interests, aspirations, spending habits.



Intellectual Property

‘Intellectual Property Rights (IPR) are like any other property right. They allow creators, or owners, of patents, trademarks or copyrighted works to benefit from their own work or investment in a creation.’

World Intellectual Property Organisation

Purpose of IP Protection Laws

- ‘Enable people to earn recognition or financial benefit from what they invent or create
- Aims to foster an environment in which creativity and innovation can flourish.’

World Intellectual Property Organisation

<https://www.wipo.int/about-ip/en/>

Types of intellectual property rights

- Patents
- Trade marks
- Registered or unregistered design
- Domain names
- Copyright
- Confidential information
- Trade secrets
- Database rights
- Your own knowledge!

Further information:

https://www.crackingideas.com/third_party/IP+for+Research

Licensing intellectual property (IP)



A licence is an agreement between you as the IP right owner and another party.

It grants them permission to do something that would be an infringement of the rights without the licence.

IP can be “licensed-out” or “licensed-in”.

You can “license-out” to another company in return for a fee.

You can “license-in” if you want to use another company’s IP to develop your own business and products.

<https://www.gov.uk/guidance/licensing-intellectual-property>

Benefits of licensing intellectual property



- Revenue generation
- Collaboration
- Increasing market penetration
- Reducing costs
- Saving time
- Accessing expertise
- Obtaining competitive advantage

Get advice from your university
if you think are creating intellectual
property that you can licence.

<https://www.gov.uk/guidance/licensing-intellectual-property>

Social enterprise

Social enterprises are businesses trading for social and environmental purposes.

A social enterprise does...

- Make its money from selling goods and services
- Cover its own costs in the long-term (though like any business, it may need help to get started)
- Put at least half of any profits back into making a difference
- Pay reasonable salaries to its staff

A social enterprise does not...

- Exist to make profits for shareholders
- Exist to make its owners very wealthy
- Rely on volunteering, grants or donations to stay afloat in the long-term (though again, it may need this sort of help to get started)

[Social Enterprise Explained, Social Enterprise UK, October 2012](#)

Influencing policy

What is a policy?

- 'A policy is a set of principles to guide actions in order to achieve a goal.
- A 'government policy', therefore describes an objective or course of action planned by the Government on a particular subject.
- Policies are usually developed by a Government Department, in order to achieve their objectives.
- Can be local, regional, national or international.

'An introduction to policymaking in the UK - Policy Guide 1', British Ecological Society, May 2017

Who are policymakers?

- “Policymaker’ is a broad term, encompassing all people involved in formulating, developing or amending policy.
- Examples include: Ministers of Government, their advisors, civil servants, Chief Scientific Advisors.

‘An introduction to policymaking in the UK - Policy Guide 1’, British Ecological Society, May 2017

How to write a policy brief

Audience

- Who is your **audience**? What is your audience like?
- Policymakers are busy people. Be **clear and concise**.
- Why should they care? Explain **why the topic is important**.
- Focus on the **impacts to people** as well as the science. Who is affected? What are the cost implications?
- **Connect the topic to local or regional areas** that they are likely to be interested in.

<https://www.parliament.uk/mps-lords-and-offices/offices/bicameral/post/about-post/writing-a-policy-brief/>

How to write a policy brief

Structure

- Keep it **short and simple** - have a clear structure
- **Make it easy to scan** by using heading and sub-headings to break up large blocks of text.
- Start with an **overview** that outlines the key points of the briefing.
- Use **figures, charts or diagrams** where suitable to help your brief be more eye-catching and appealing.

<https://www.parliament.uk/mps-lords-and-offices/offices/bicameral/post/about-post/writing-a-policy-brief/>

How to write a policy brief

Content

- Ensure that the brief is clear about how the issue you are writing about **relates to policy**.
- What is your brief is trying to achieve? Shape the content accordingly. What are the actionable messages / **recommendations with timescales**?
- **Present the evidence** for your argument.
- Be explicit about methodological **limitations** and / or the strength of the evidence you are presenting.

<https://www.parliament.uk/mps-lords-and-offices/offices/bicameral/post/about-post/writing-a-policy-brief/>

How to write a policy brief

Content

- Include **handy facts and figures** they can make use of.
- Avoid emotive language and **let the research speak for itself**.
- **Don't assume prior knowledge**. Minimise jargon and acronyms.
- Highlight consensus and ongoing debate. Be clear about how you are building on existing knowledge.
- Be clear about where there is **uncertainty**.
- Provide your contact details and make sure your brief is dated.

<https://www.parliament.uk/mps-lords-and-offices/offices/bicameral/post/about-post/writing-a-policy-brief/>

Public engagement

Research

Applied
Theoretical
Collaborative
Co-produced research

Social Media

Blogs & microblogging
Online forums & discussions
Wisdom of the crowd
Youtube lectures & demos

Info-tainment

Media
Festivals
National events

Knowledge Transfer

Outreach
Public lectures
Research dissemination

What is public engagement?

Learning

Professional development
Lifelong learning
Networking & sharing
Schools liaison
Widening participation

Knowledge Exchange

Influencing policy

Community

Student & staff volunteering
Cultural & social partnerships
Opening up spaces/facilities

Planning public engagement activity

Know what your aims are

Consider whether you want to:

- **'inform** people and make your research more accessible
- **consult** people and explore the ethical and social implications of your research
- **collaborate** with people and share expertise'.

<https://wellcome.ac.uk/grant-funding/guidance/planning-your-public-engagement>

Wellcome Trust step by step guide to planning public engagement activity

1. Planning

2. Resourcing

3. Piloting

4. On-going development

5. Final evaluation

6. Wider dissemination

Planning your public engagement activities: Step by step guide

	<i>Audience</i>	<i>Methods</i>	<i>Collaboration</i>	<i>(Evaluation) Impact on Audience</i>	<i>(Evaluation) Impact on Research</i>
1. Planning	Consider the main stakeholders: Which public groups or organisations are most likely to be interested in your research?	Consider the best ways to engage them: What methods, activities, formats or locations are most useful?	Consider partners who are engaging your audiences already e.g. patient or youth groups, charities, museums, arts orgs or broadcasters	Consider the intended outcomes: What will everyone involved gain from participating and how will you evaluate this?	Consider integration with your research plans: How could this add to your research and benefit you and your colleagues?
2. Resourcing	Be realistic and choose one or two specific audience groups or stakeholders to work with	Consider staff time and applying for funding. Does your team need additional training or support from your institution's public engagement staff?	Consider your partners/ collaborators' resource needs and how they could benefit from being involved	Consider commissioning an external evaluator to ensure a robust evaluation. Do you need additional funding for this?	Consider commissioning an external evaluator to ensure a robust evaluation. Do you need additional funding for this?
3. Piloting	Did your pilot reach your intended audiences effectively? Has it revealed other audiences you would now like to consider?	Could you adapt your methods, activities, formats or locations to better suit your audience? Would different methods be better?	Consider what you've learnt from beginning to work with your partner: How can you build on the successes of your collaboration?	Conduct a formative evaluation: Were there any unexpected outcomes? What needs to change to improve your activities?	Reflect on your research plans: What aspects could be changed to ensure maximum benefit to stakeholders?
4. On-going development	Continue to work with your chosen groups, keep listening and think of secondary audiences	Keep adapting your methods and reflecting on their value	Has the feedback from partners, collaborators and participants highlighted any other potential stakeholders or interested groups?	Are the outcomes of your work continuing to meet everyone's aims and objectives? Is their mutual benefit for all parties involved?	What can you share about the progress of your research at this stage? Have new things emerged that you could build into your engagement plans?
5. Final evaluation	Reach: Consider numbers of primary and secondary audiences reached as well as unintended audiences, colleagues and peers	Quality: Did you produce high quality outputs? Did the audience enjoy participating? How could methods be improved for the future?	Impact: Was your collaborative process successful? How will you build on these relationships in the future?	Impact: Did you achieve a good depth of engagement with the science? Did you achieve your intended outcomes?	Impact: Did your engagement work impact on your research? Has it had an impact on your peers, institution or field of work?
6. Wider dissemination	Share the outputs with a wider audience: How could you broaden the reach and lay the foundations for future work?	What methods could you use to disseminate your evaluation findings more widely? How can you share your learning?	What connections and networks can your collaborators/ partners offer to share your research and engagement learning?	Could you continue to provide resources, links to further information and connect your audiences to other organisations?	Could you work with your institution's public engagement staff to share your evaluation with others inside your institution and beyond.

<https://wellcome.ac.uk/grant-funding/guidance/planning-your-public-engagement>

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Creating impact

Creating impact



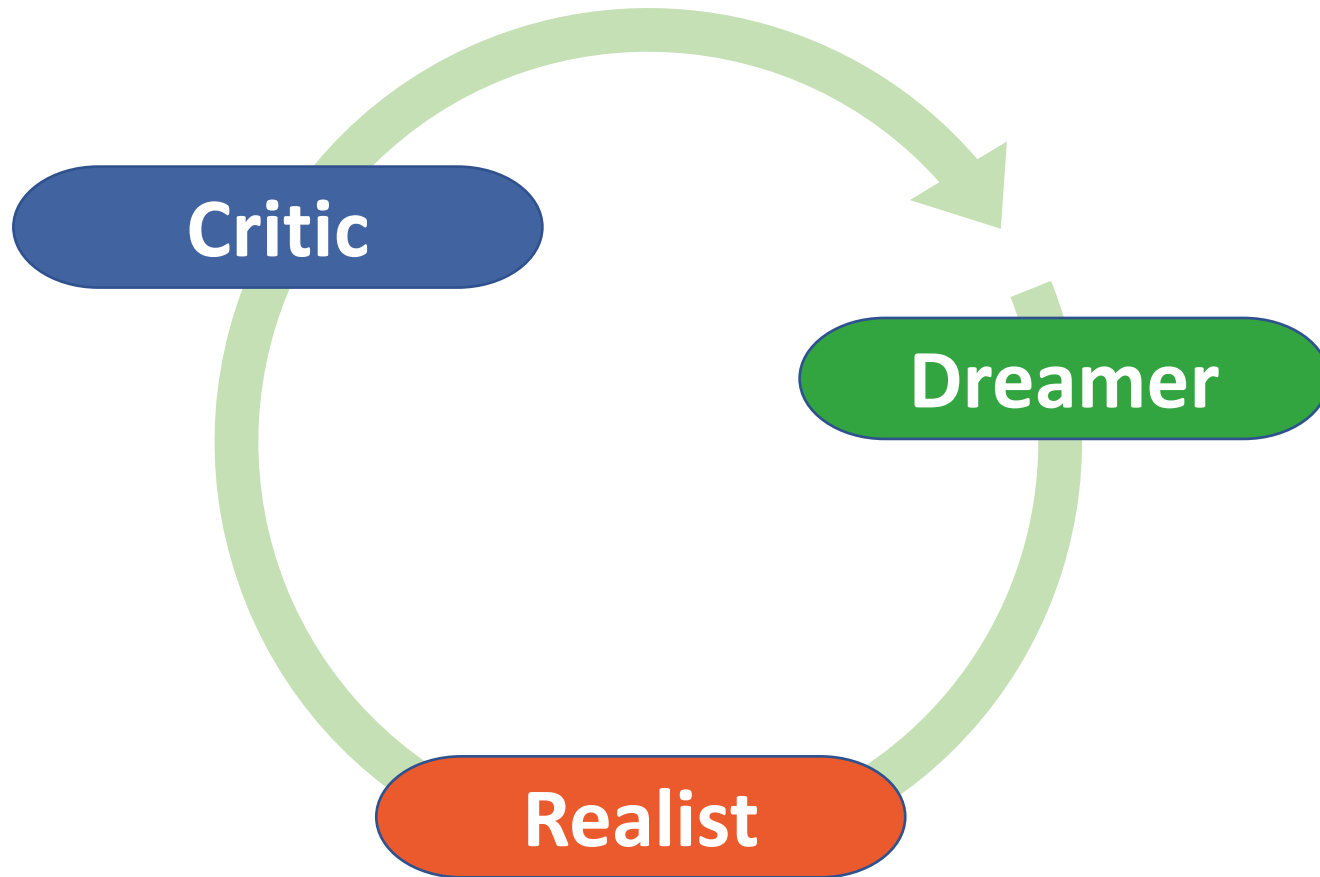
Impact can be created at any stage of your research.

Think about who you are generating the impact for.

Think about potential impacts when you write your grant proposal.

<https://www.ox.ac.uk/research/support-researchers/using-research-engage/research-impact-creating-capturing-and-evaluating>

Disney Creative Strategy



Disney Creative Strategy

Dreamer (WHAT?)

- Be open to any ideas.
- Freely associate and think outside the box.
- During this stage there are no limitations.

Realist (HOW?)

- How can you balance your ideas with the resources you have available?
- Make your creative idea happen.

Critic (WHY?)

- Look critically at your idea.
- What are its strengths and weaknesses?
- Find faults in the idea.
- Refine it.

Disney Creative Strategy – Questions to ask

Dreamer	Realist	Critic
Why do we do it?	How do we do it?	Where are the problems?
What is the purpose?	What do we need?	What can go wrong?
What are the benefits?	What are our resources?	What if ... (something happens)?
What can be done?	Who can help us?	What is missing?
What can we achieve?	How much (of those resources) do we need?	What are the risks?

Barton, M., Müllerová, J., & Svobodová, I. (2012). Disney Strategy in Management-Inspiration from Arts and Creative Industries. *Štefan Majtán a kolektív Aktuálne Problémy Podnikovej Sféry*, 15.

Evaluating impact

Types of data to use

Quantitative data - 'same questions should be used throughout the evaluation and responses gathered from a representative sample'.

Qualitative data - 'explore the participants' experience in more depth than quantitative data'.

Project team records - 'keeping an evaluation journal allows the programme team to explore and reflect on the process of developing and delivering the project'.

<https://esrc.ukri.org/research/impact-toolkit/developing-a-communications-and-impact-strategy/measuring-success/the-evaluation-process/>

Evaluation process

1. **Aim** (what do you want to achieve? Big picture!)
2. **Objectives** (what you need to do to achieve your aim?)
3. **Evaluation questions** (what do you want to know?)
4. **Methodology** (what strategy will you use?)
5. **Data collection** (what techniques will you use to collect your evidence?)
6. **Data analysis** (how will you analyse your data?)
7. **Reporting** (who will be reading your report?)

Future Planning – Steps for successful impact

- Envision your potential impact.
- Identify and define targeted stakeholders.
- Get specific about your impact goals and make them measurable. How will you monitor and evaluate your progress and success?
- Check you have relevant activities to reach each of your goals and to meet the needs and interests of your stakeholders.
- Check any assumptions you may be making and consider potential barriers that you may need to manage.
- Create a communications plan that supports your plans for impact.

Consider how this experience might enhance your career and the careers of those you lead.

Reflection activity

- creating your own impact

- What kind of research impact activity can you develop from your own research? (Or list what you are already doing.)
- What skills and knowledge do you need to help you develop this activity?
- What resources will you need?

Questions ???



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29 September 2020

